SOVEREIGN GOVERNMENT CLOUD

CSIR Secure Web Appliance OS (CSIR-SWA OS)

Next-Generation Hardened, Immutable, and Containerized Web Hosting Appliance

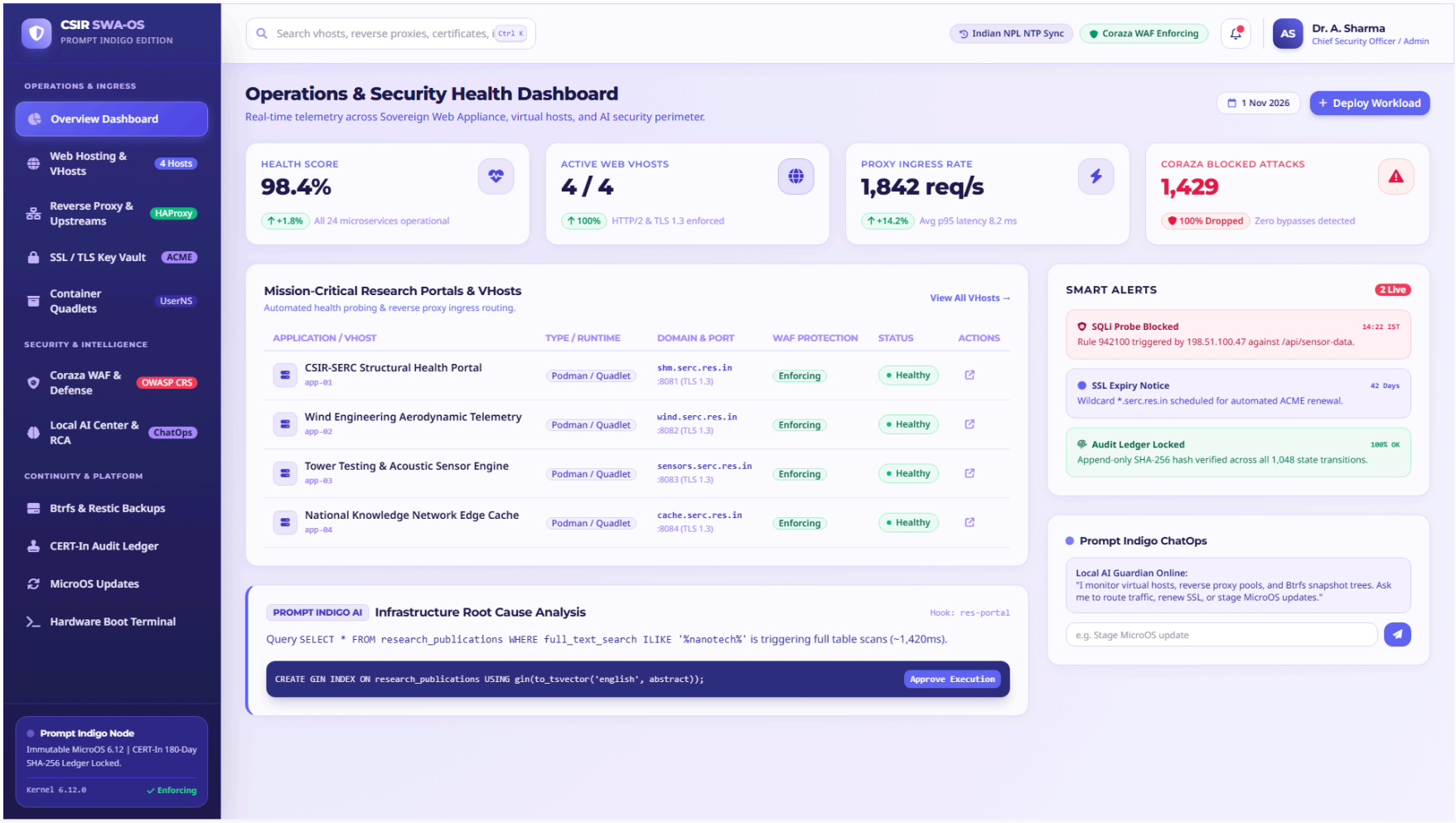
CORE ARCHITECTURAL MANDATE

- Engineered for classified research laboratories, government web portals, and critical public infrastructure.
- Prompt Indigo AI SaaS Architecture with Tailwind CSS, daisyUI, and Glassmorphic aesthetics.
- Integrated Multi-Tenant Web Hosting, Virtual Host Manager, HAProxy Reverse Proxy, and ACME SSL Key Vault.

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 1: Central Operations & Security Health Overview Dashboard

Dedicated Verification Slide



Executive Health KPI
98.4% operational health score across 24 microservices

Proxy Ingress Telemetry
1,842 req/s throughput with sub-10ms p95 latency

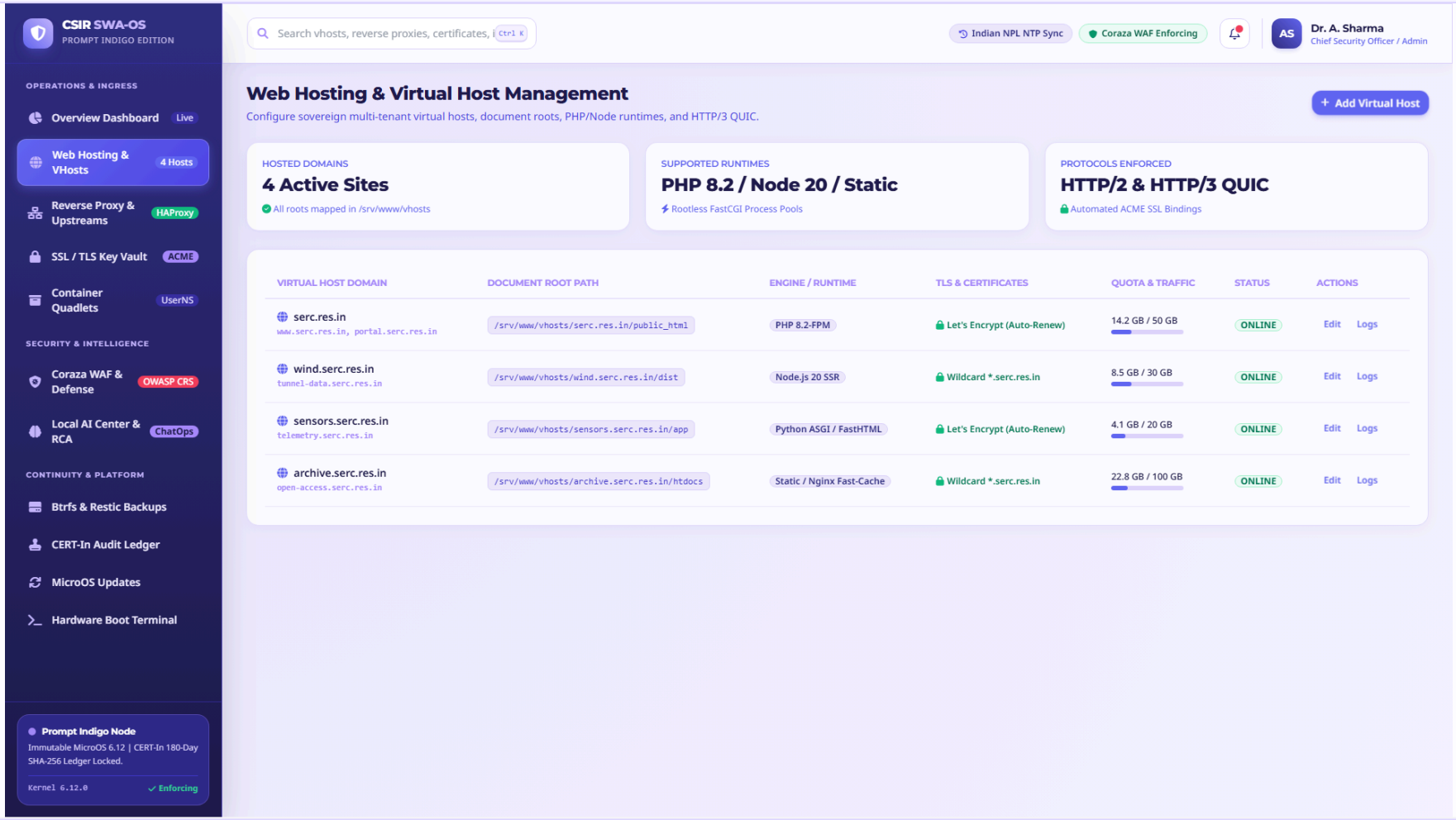
Coraza Threat Shield
1,429 blocked attacks with 100% drop & zero bypasses

Prompt Indigo ChatOps
Real-time security timeline with sovereign AI assistant

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 2: Native Web Hosting & Multi-Tenant Virtual Host Management

Dedicated Verification Slide



Multi-Tenant Isolation
Enforced document roots under /srv/www/vhosts with strict POSIX perms

Polyglot Runtimes
Native support for PHP 8.2-FPM, Node.js 20 SSR, Python ASGI & Static

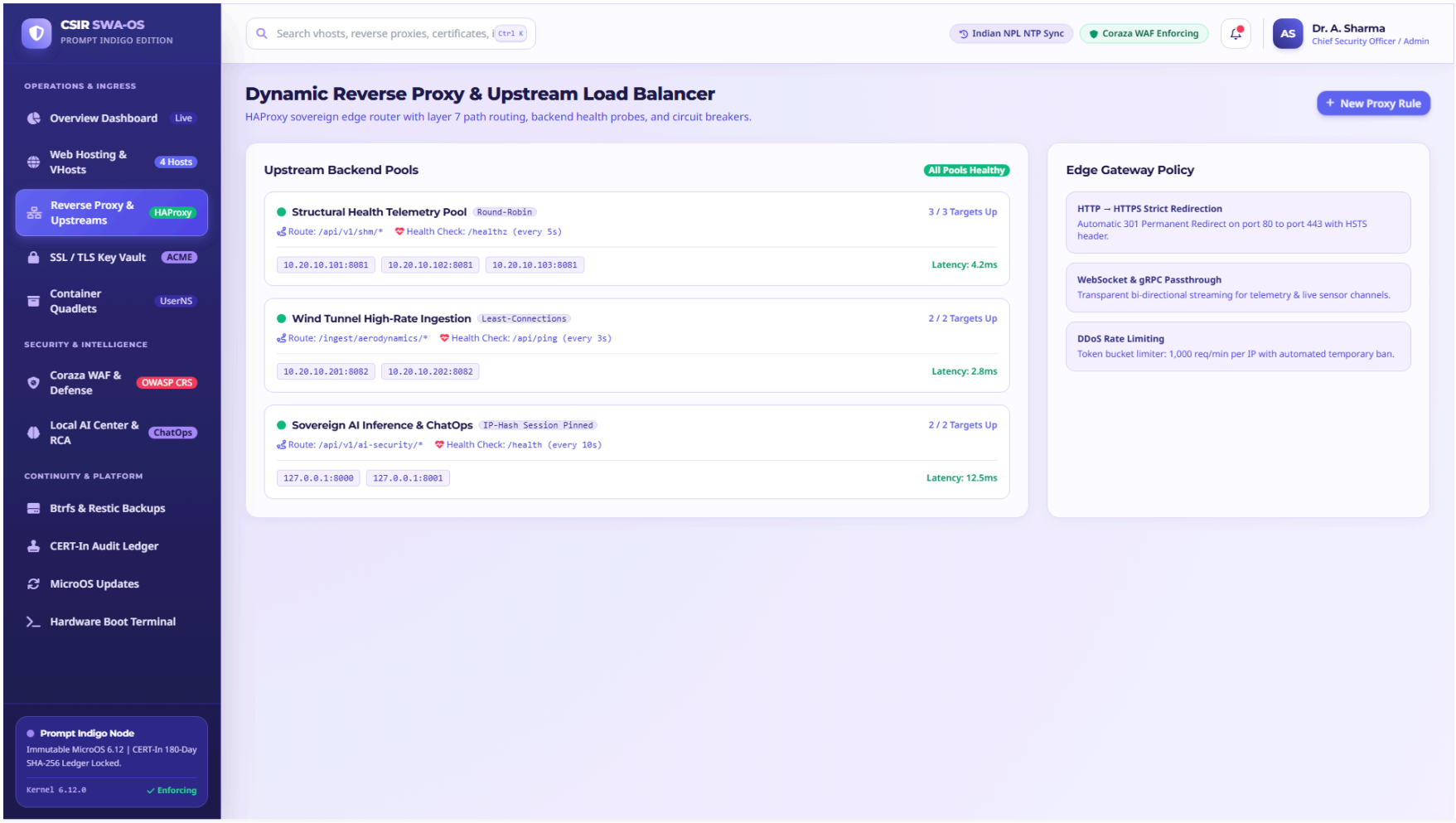
Storage Quota Bars
Visual disk consumption tracking with automated threshold enforcement

Active SSL Binding
Instant binding to automated Let's Encrypt and wildcard certificates

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 3: HAProxy Dynamic Reverse Proxy & Upstream Server Pools

Dedicated Verification Slide



Upstream Target Pools
Dedicated server pools for Structural Health, Wind Tunnel, and Local AI

Balancing Algorithms
Round-Robin, Least-Connections, and IP-Hash session pinning

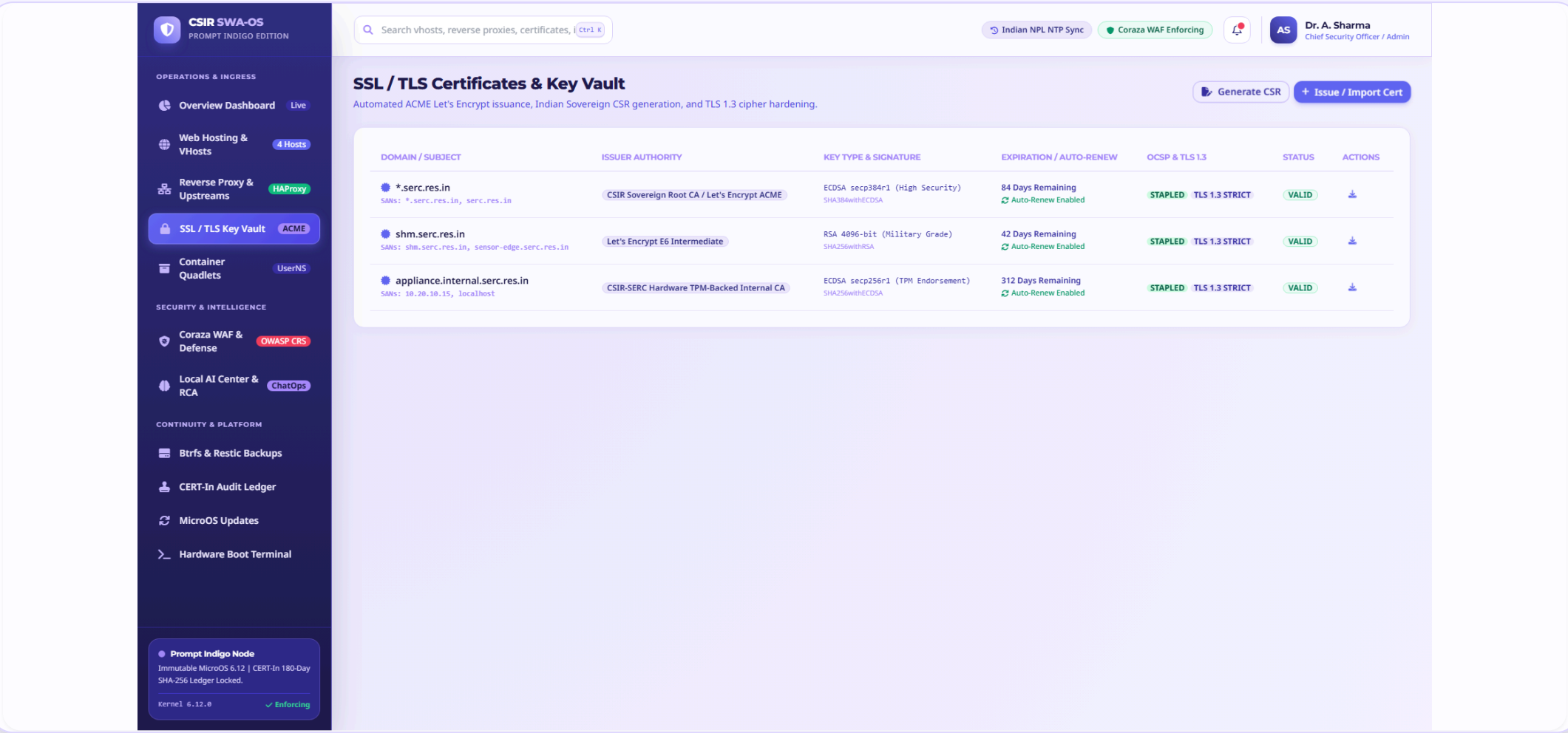
Active Health Checks
Deterministic HTTP probes every 3-10s with automatic failover

Edge Hardening
Strict HTTP-to-HTTPS redirect, WebSocket passthrough, and DDoS rate limits

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 4: Sovereign SSL / TLS Certificate Vault & ACME Key Management

Dedicated Verification Slide



Automated ACME Lifecycle
Zero-touch renewal for *.serc.res.in 30 days prior to certificate expiry

Military-Grade Keys
ECDSA secp384r1 and RSA 4096-bit institutional private keys

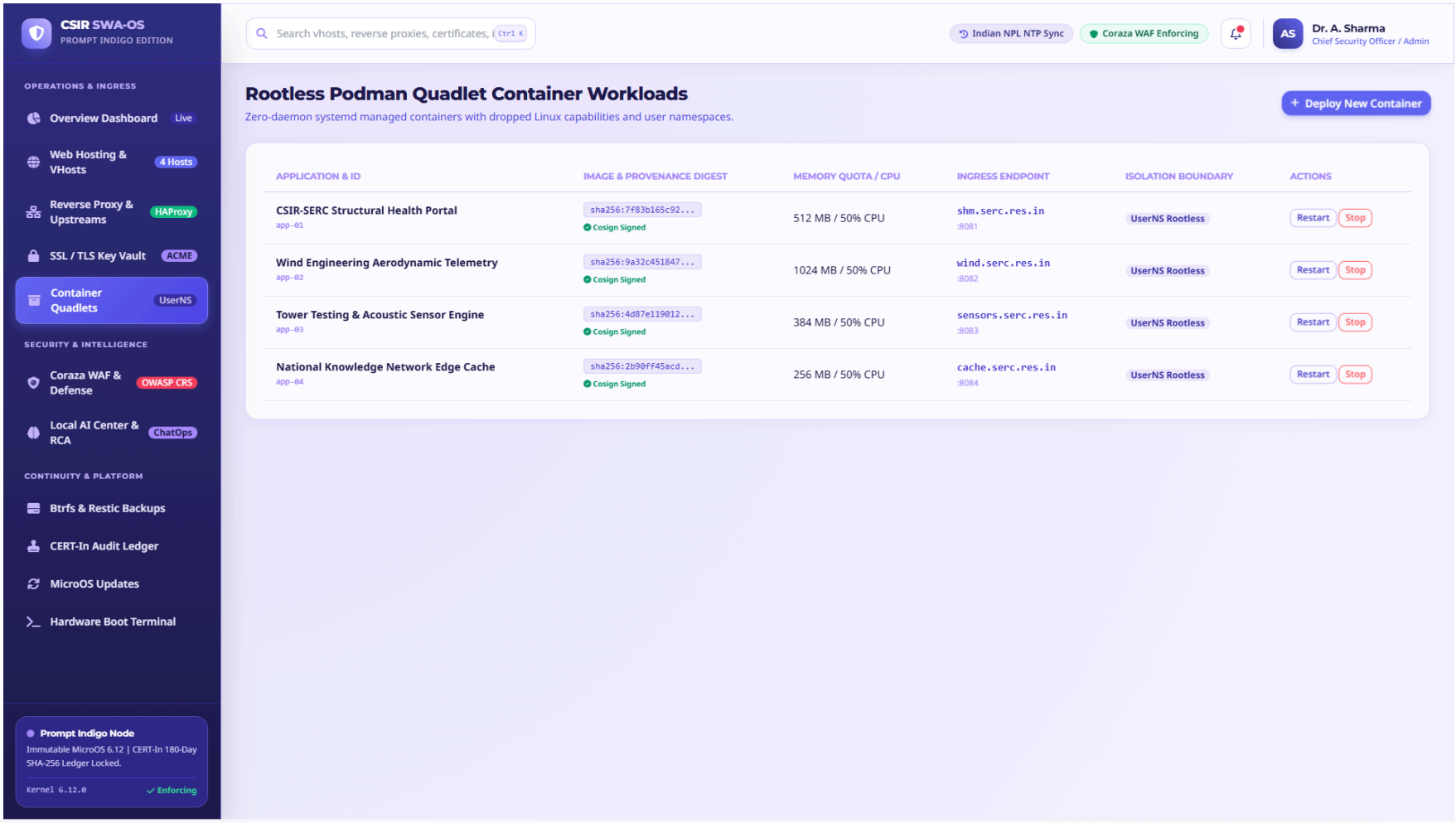
Strict TLS 1.3 Suite
Hardened cipher suites with zero legacy fallback vulnerabilities

Sovereign CSR Tool
Integrated generation of certified government certificate signing requests

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 5: Rootless Podman Quadlet Container Workloads

Dedicated Verification Slide



Cosign Verification
Mandatory cryptographic signature verification before workload ignition

Rootless UserNS Sandbox
Zero root privileges inside containers, mitigating container breakout risks

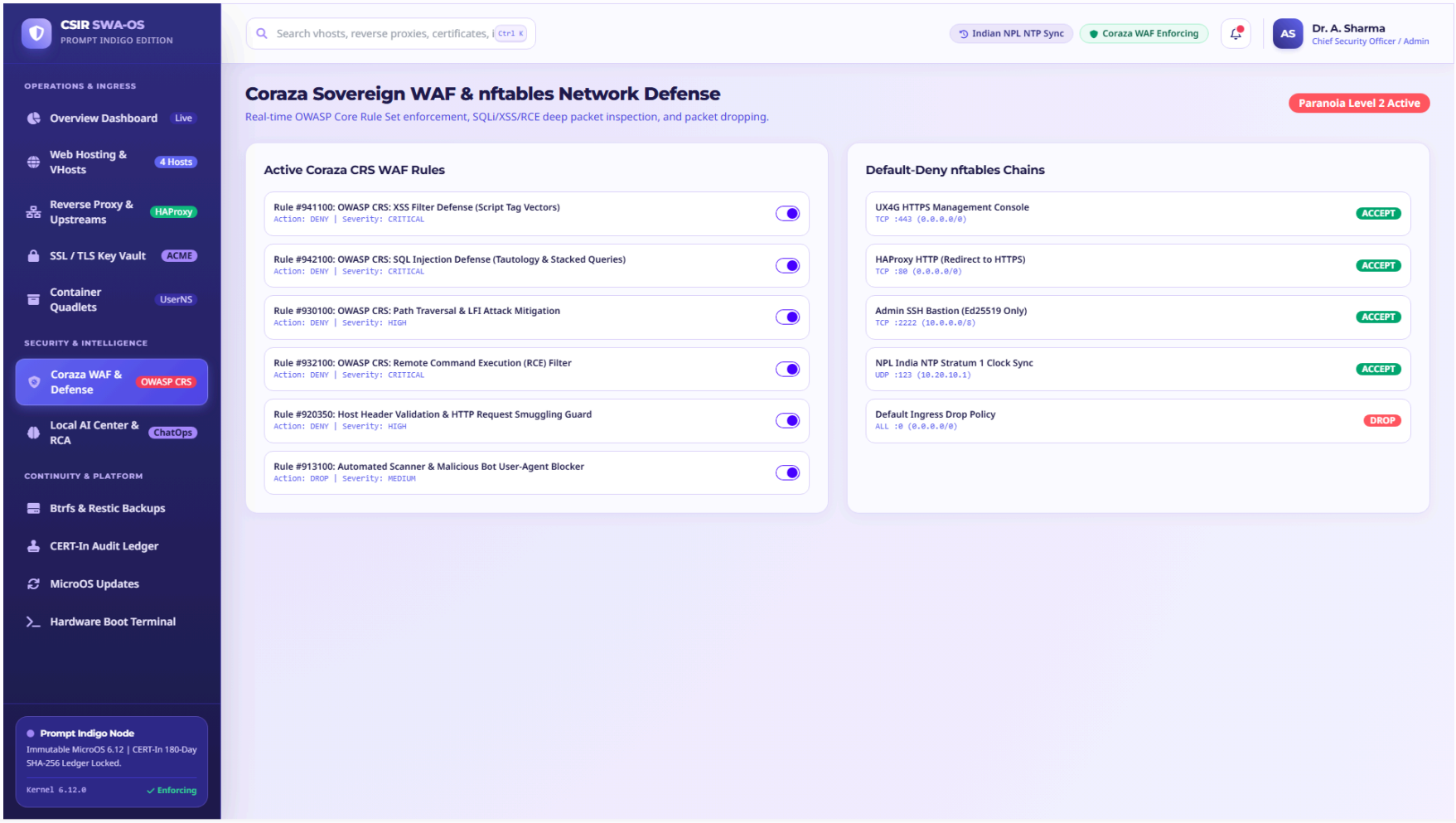
Strict Hard Quotas
Enforced memory allocations (256MB-1024MB) and 50% CPU ceilings

Systemd Supervision
Automated health watchdog monitoring with instant restart on crash

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 6: Coraza Sovereign WAF & nftables Network Defense

Dedicated Verification Slide



OWASP CRS v4 Defense
Active mitigation of SQLi (942100), XSS (941100), RCE (932100) & LFI (930100)

Paranoia Strictness L2
Enhanced scrutiny on administrative endpoints without disrupting users

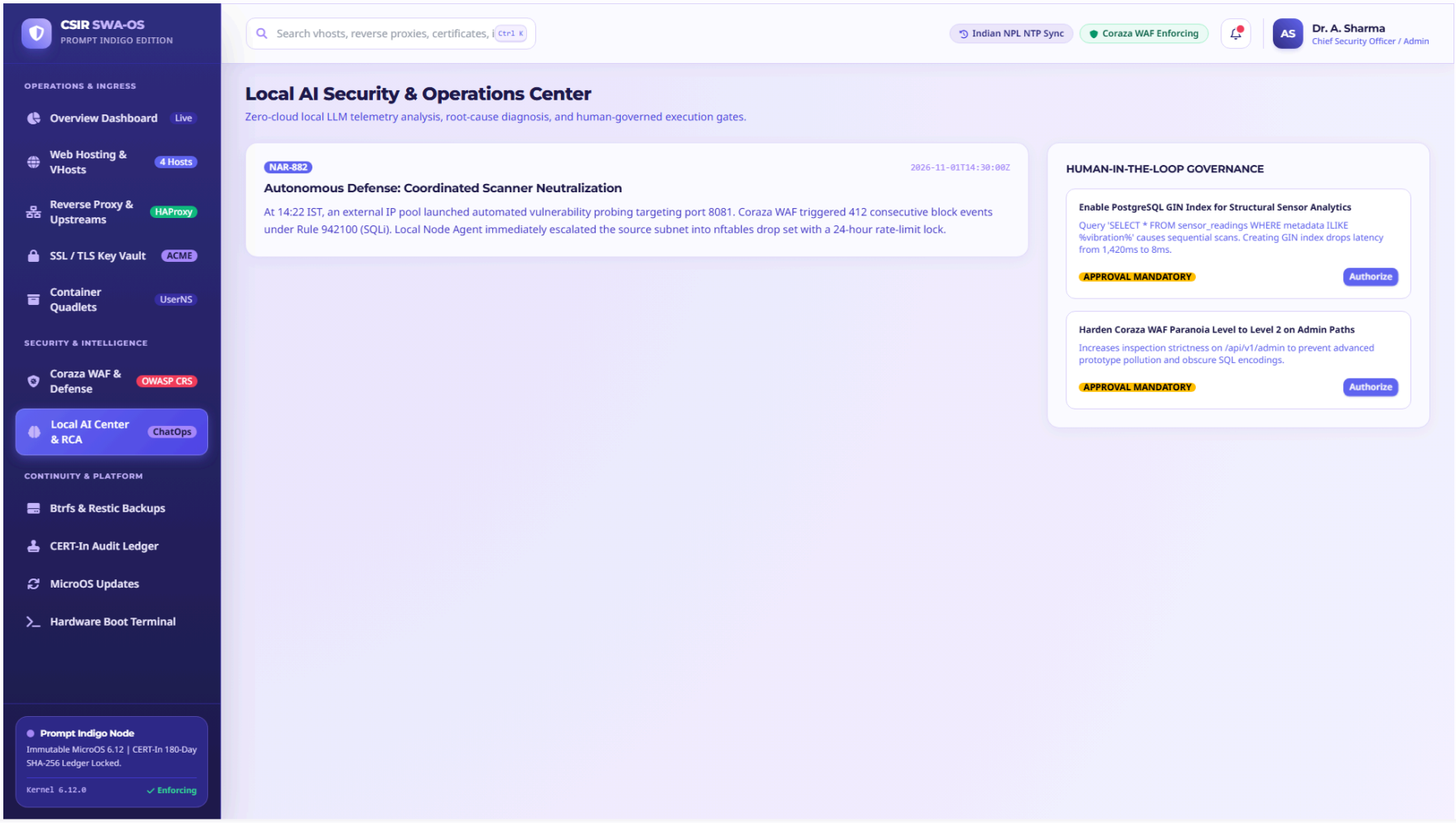
nftables Kernel Filter
Default-deny ingress policy with strict whitelisting for bastion SSH

Zero Cloud Telemetry
100% on-premises packet inspection preserving national confidentiality

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 7: Local AI Security Center & Incident Root Cause Analysis

Dedicated Verification Slide

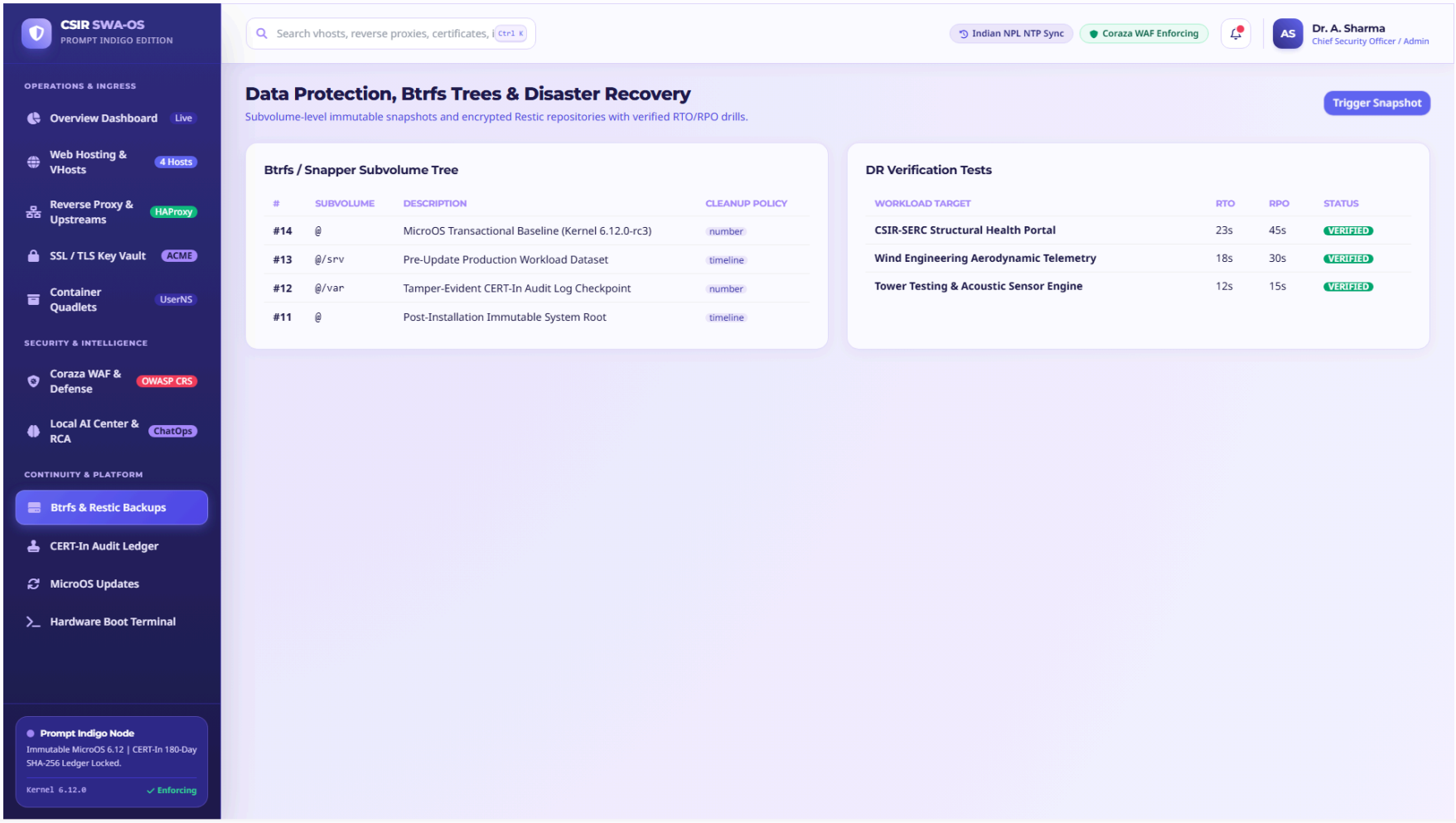


- **Incident Synthesis**
  Autonomous plain-English briefings of scanner probes (Narrative NAR-882)
- **Human Approval Gates**
  Strict mandatory human approval required before AI mitigations are committed
- **Database Diagnostics**
  Automated RCA identifying unindexed queries and optimal connection pooling
- **Sovereign ChatOps**
  Natural language query interface for instant cluster operational control

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 8: Immutable Data Protection & Btrfs Automated DR Drills

Dedicated Verification Slide



Btrfs Subvolume Tree
Granular snapshot management across root (@), data (@/srv), and logs (@/var)

Verified DR Drills
Measured RTO (12s-23s) and measured RPO (15s-45s) for production portals

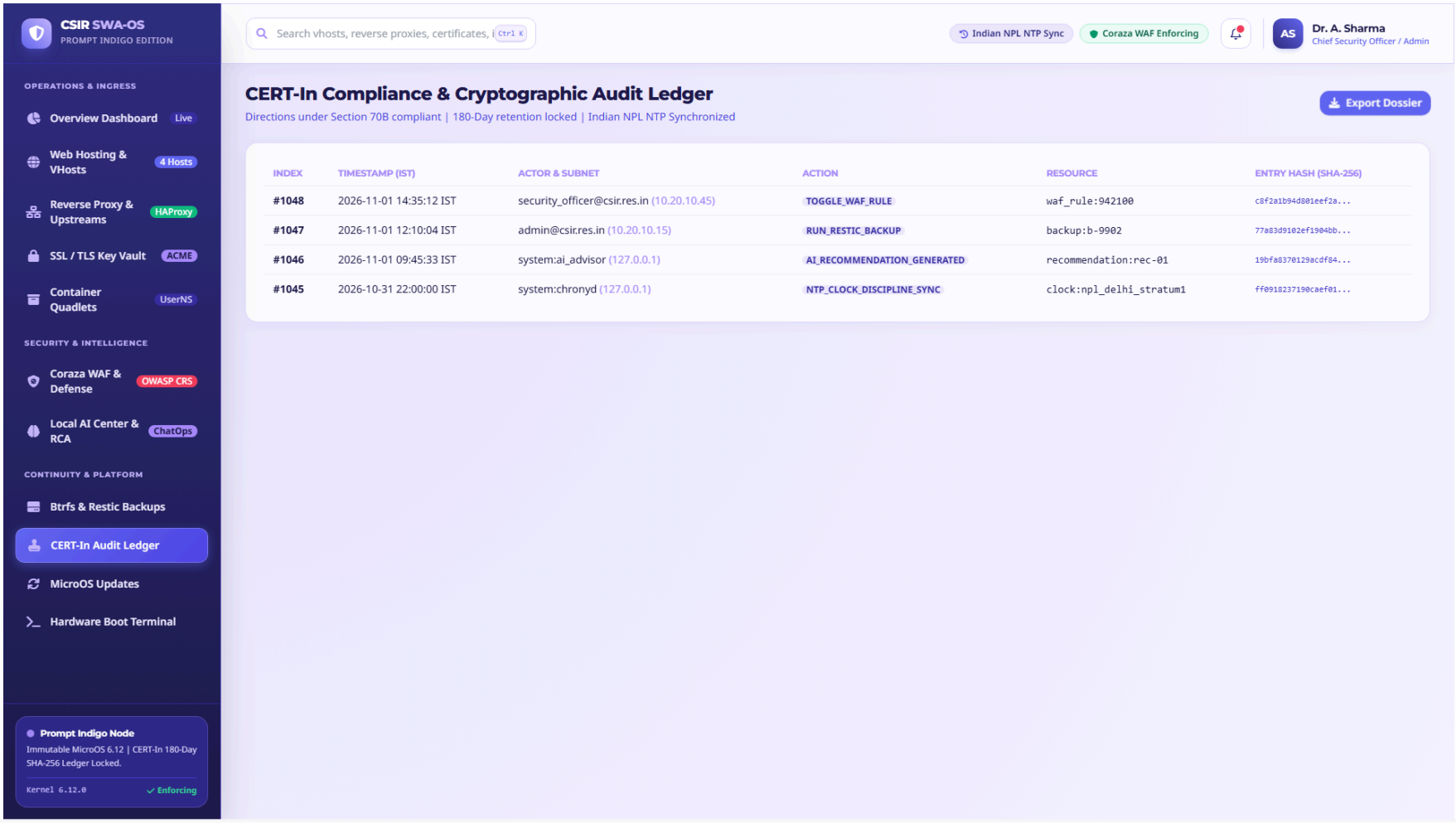
SHA-256 Hash Auditing
Full cryptographic verification confirming zero data degradation post-restore

Ransomware Immunity
Read-only snapshot hierarchies guarantee instant rollback against crypto-malware

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 9: CERT-In Section 70B Cryptographic SHA-256 Audit Ledger

Dedicated Verification Slide



SHA-256 Hash Chaining
SHA256(prev_hash | entry_hash) guarantees mathematical tamper-evidence

Section 70B Compliance
Comprehensive auditing of every privileged state change and WAF toggle

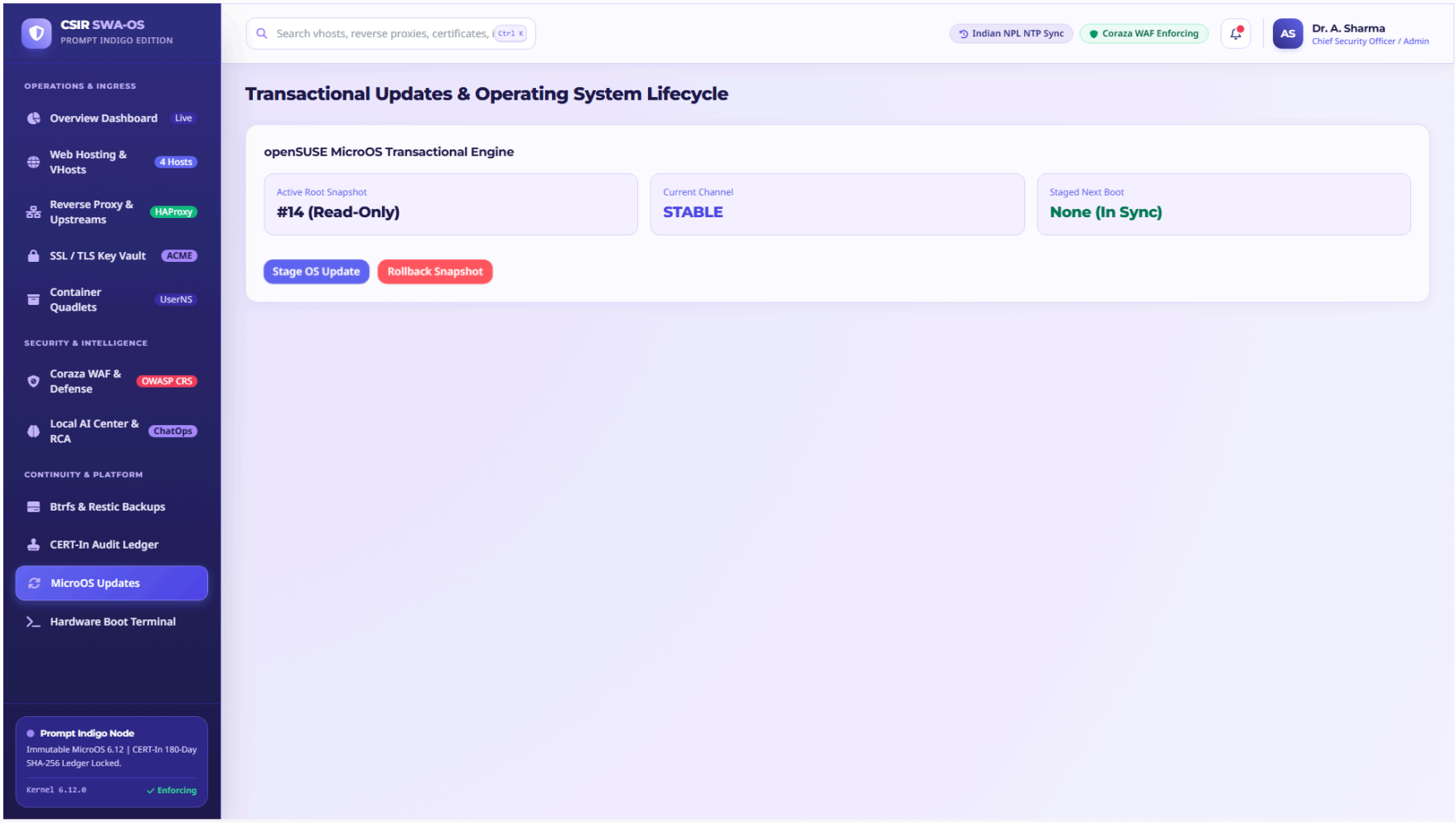
NPL Stratum 1 Discipline
Chronyd synchronized strictly against Indian National Physical Laboratory

1-Click Certified Dossier
Rapid statutory export fulfilling CERT-In 6-hour incident window

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 10: openSUSE MicroOS Transactional-Updates & Rollback

Dedicated Verification Slide



Read-Only System Root
Appliance operating system executes from immutable snapshot #14

Atomic Update Staging
OS updates install into isolated snapshots with zero runtime interruption

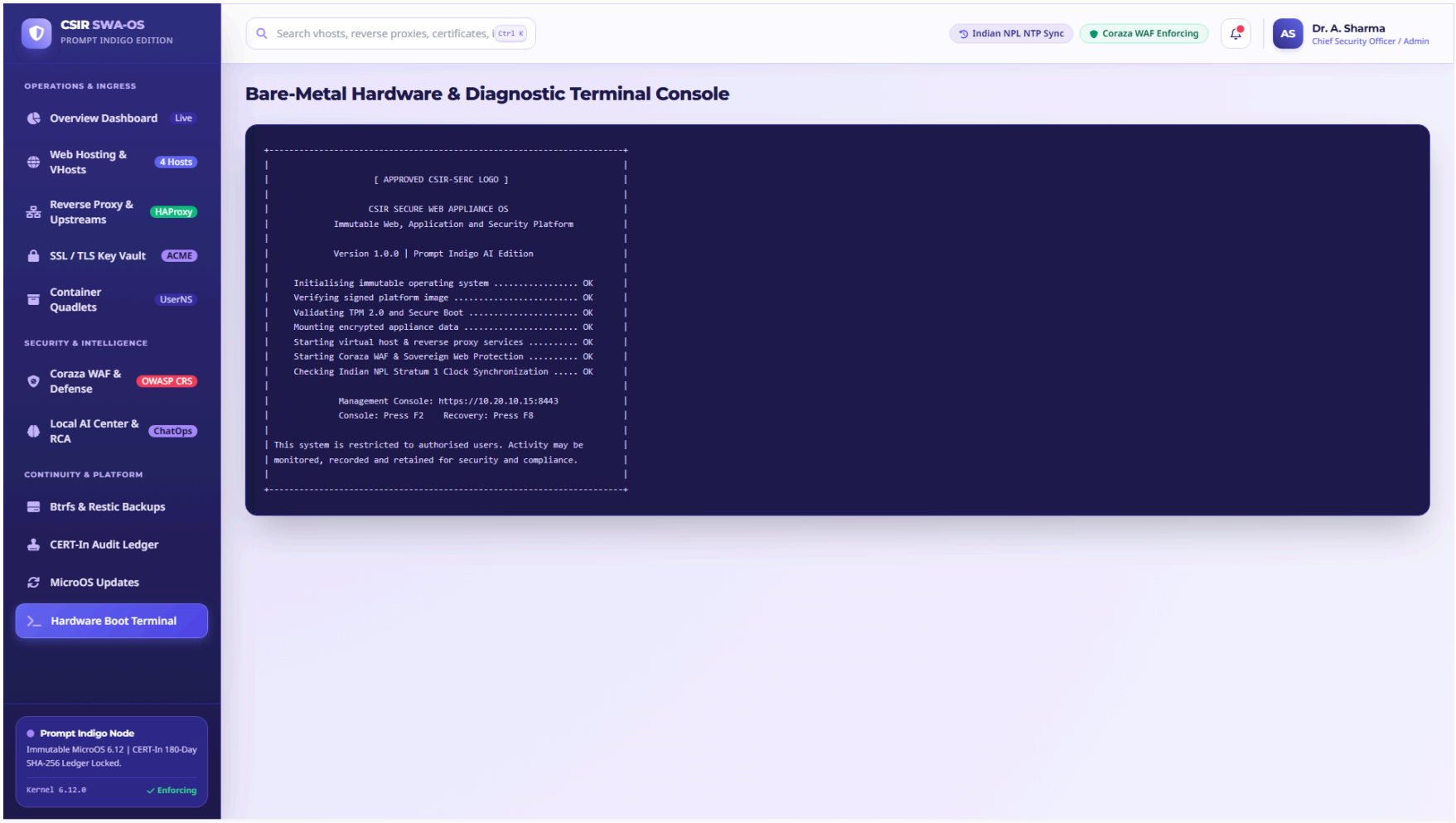
Health-Gated Reversion
Automated rollback reboot if post-boot health probes detect failure

1-Click Instant Rollback
Administrative console allows immediate reversion to known-good baseline

VISUAL QUALIFICATION: HARDWARE CONSOLE SUBSYSTEM

Figure 11: Bare-Metal Hardware & Diagnostic Terminal Console

Dedicated Verification Slide



- **Diagnostic Boot Banner**
  Official ASCII institutional console banner displayed on bare-metal boot
- **Hardware Validation**
  Automated validation of TPM 2.0, Secure Boot, LUKS disks & NPL clock sync
- **Emergency Recovery Keys**
  F2 hotkey for console access, F8 hotkey for emergency immutable rollback
- **Statutory Access Lock**
  Legal access warning restricting appliance entry to authorized personnel

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 12: High-Resolution GRUB2 Bootloader & Deployment Menu

Dedicated Verification Slide

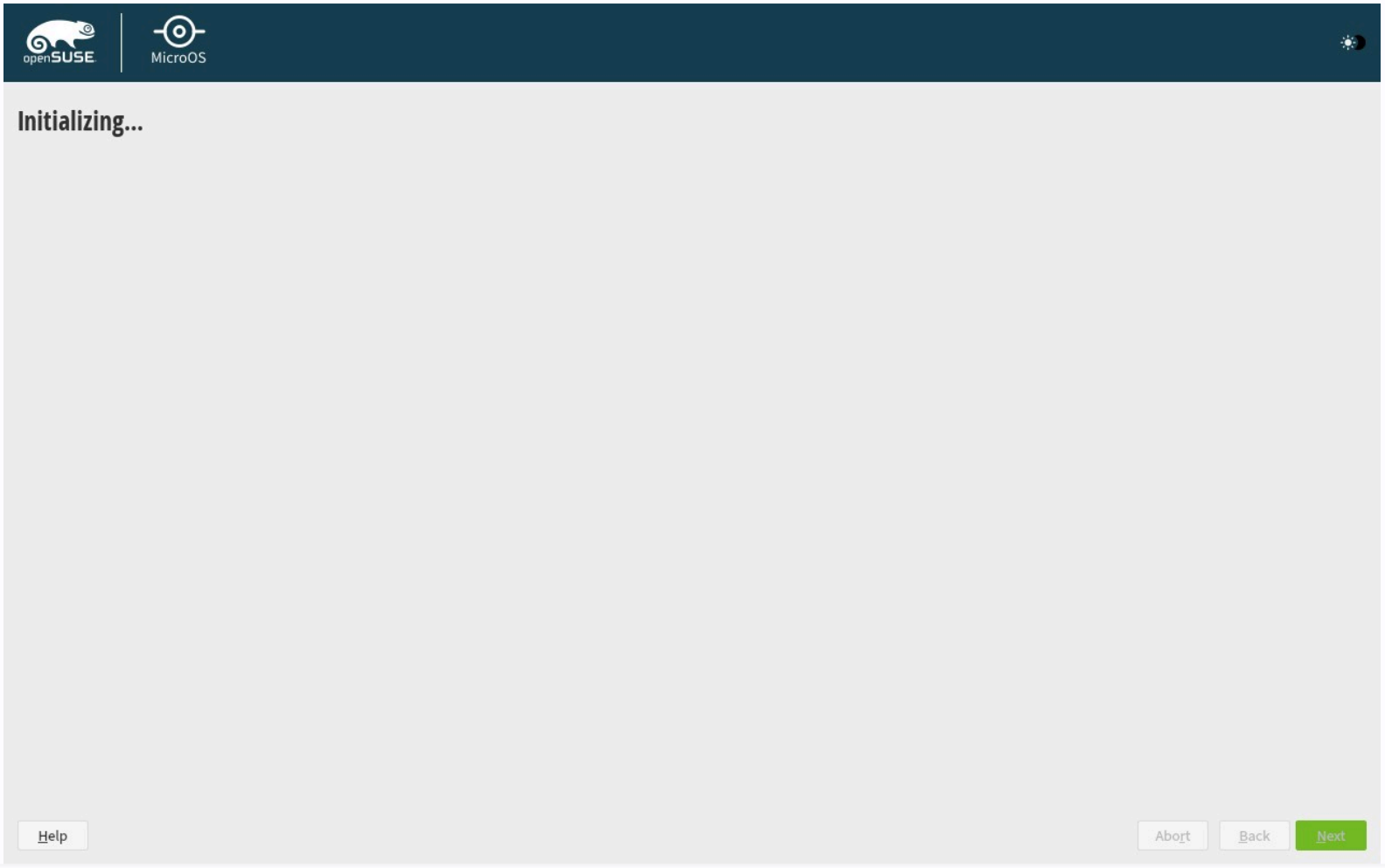


- **1024x768 Graphical Canvas**
  TrueColor graphical frame-buffer with official CSIR-SERC institutional seal
- **Turn-Key Boot Modes**
  Automated Express Install, Easy Interactive Install, Memtest & Hardware Probe
- **Dual Architecture**
  Full support for UEFI Secure Boot (x86_64-efi) and legacy BIOS hardware
- **Offline Monolithic DVD**
  Self-contained ISO with 100% offline packages and zero internet requirement

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 13: Graphical YaST Installer Engine & Combustion Automation

Dedicated Verification Slide



Combustion Engine
Automated execution of first-boot appliance hardening scripts

Hardware RAM Probing
Automated detection allocating memory for graphical and container workloads

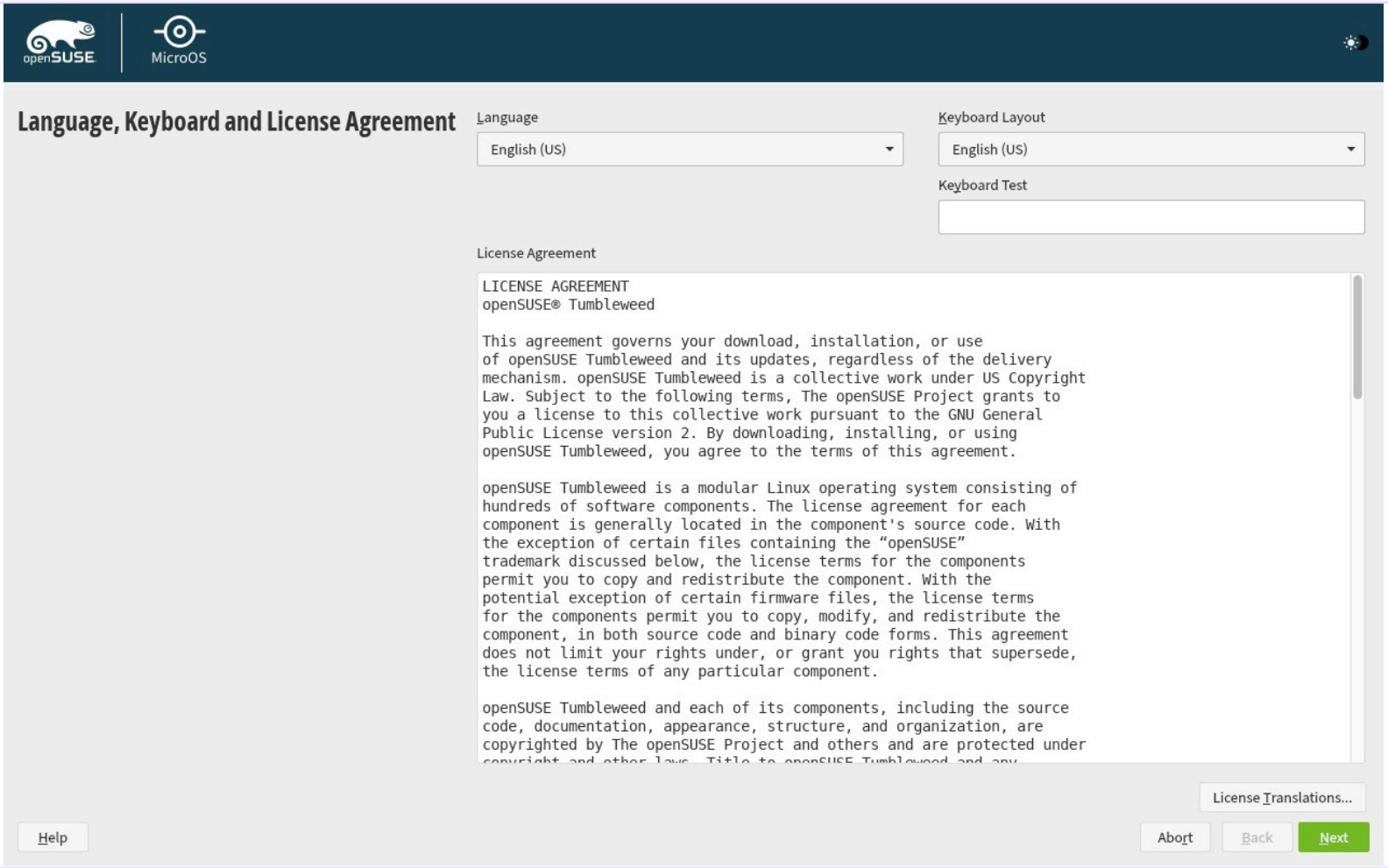
Hardened Base Profile
openSUSE MicroOS transactional minimal profile with zero bloatware

Package Verification
All installer RPM packages verified against CSIR institutional GPG key

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 14: Automated Network Interface Discovery & Time Synchronization

Dedicated Verification Slide

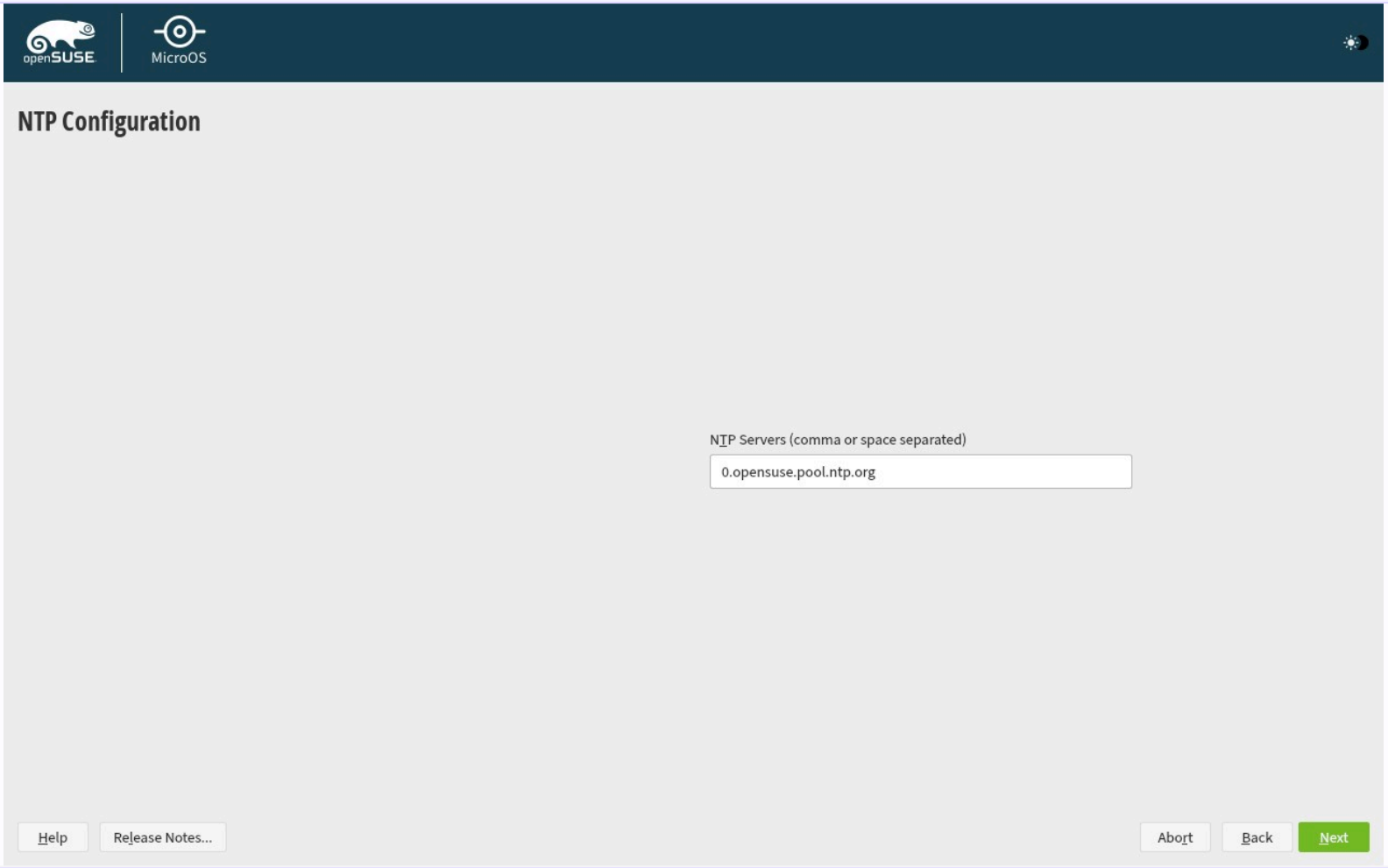


- **Interface Probing**
  Deterministic identification of 10G/1G physical Ethernet and virtual NICs
- **Dual-Stack Addressing**
  IPv4 and IPv6 automatic acquisition with static management override
- **Sovereign Clock Sync**
  Chrony disciplined directly by Indian National Physical Laboratory Stratum 1
- **Default-Deny Ingress**
  Firewall rules primed and active prior to network link activation

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 15: Immutable Btrfs Subvolume Layout Scheme (@, @/var, @/srv)

Dedicated Verification Slide



Read-Only Root (@)
Mounted strictly read-only to ensure complete malware and tampering immunity

Data Subvolume (@/srv)
Segregated read-write subvolume for virtual hosts, web roots & Podman

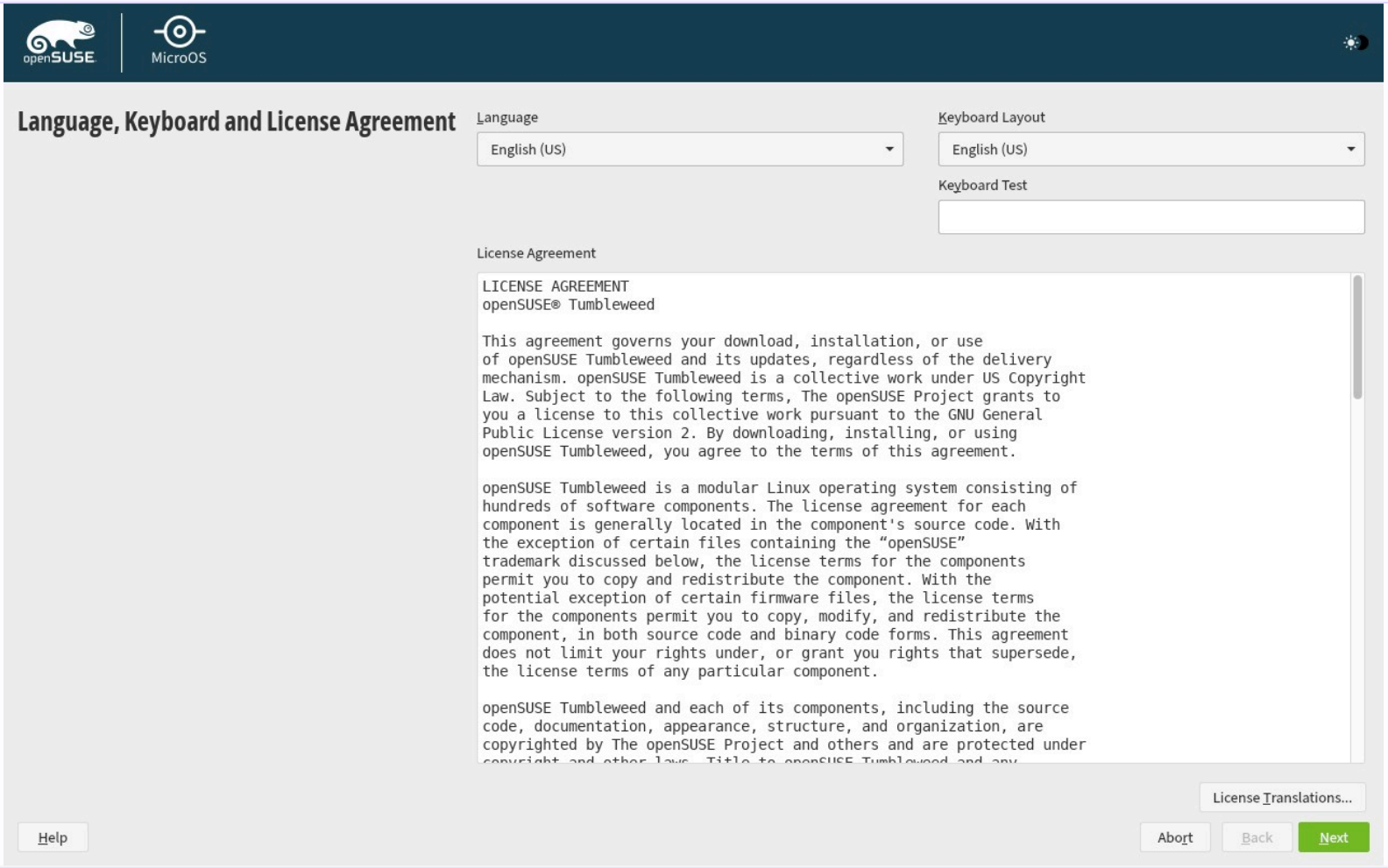
Audit Subvolume (@/var)
Dedicated subvolume for CERT-In 180-day append-only cryptographic logs

Independent Snapper Trees
Granular snapshot retention policies preventing log churn from polluting root

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 16: Final Installation Confirmation & Sovereign Payload Commitment

Dedicated Verification Slide



Full Appliance Payload
Complete offline provisioning of Coraza WAF, HAProxy, Podman & Vue 3 SPA

Security Hardening
Disabling non-essential daemons, SHA-512 password hashing & SSH bastions

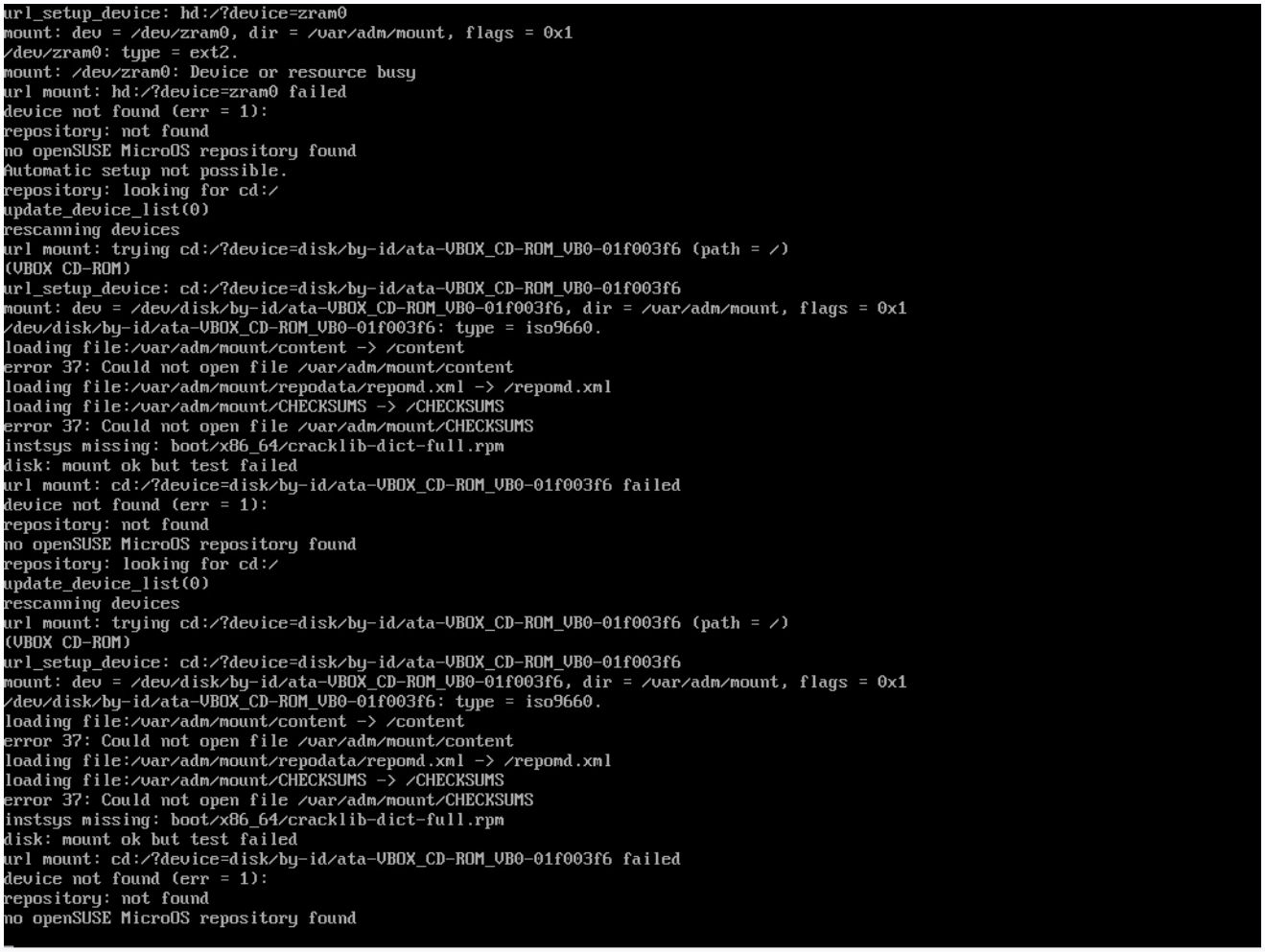
Storage Commitment
Atomic Btrfs formatting with initial baseline snapshot generation

Secure Bootloader
Digitally signed GRUB2 committed to EFI System Partition (ESP)

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 17: Diagnostic Linuxrc Installation Console & TTY3 Shell

Dedicated Verification Slide



Kernel Telemetry
Live tracking of device driver probing, hardware interrupts & storage mounting

Storage Attachment
SATA/NVMe controller discovery and LUKS encrypted container initialization

Diagnostic Audit Shell
Unprivileged audit terminal accessible via Ctrl+Alt+F3 during installation

Fail-Safe Logging
Direct logging of all low-level hardware interactions for post-install audit

ARCHITECTURAL SYNTHESIS

Comprehensive Defense-in-Depth & Deployment Summary

Four Pillars of Sovereign Security

- **Layer 1 - Host Immutability:** Read-only Btrfs root snapshot prevents malware foothold persistence.
- **Layer 2 - Workload Sandboxing:** Rootless Podman Quadlets with UserNS isolation eliminate privilege escalation.
- **Layer 3 - Real-Time WAF Defense:** Coraza OWASP CRS v4 inspects 100% on-premises with zero cloud leaks.
- **Layer 4 - Cryptographic Audit:** SHA-256 hash-chained ledger guarantees CERT-In statutory adherence.

Turn-Key Deployment Modalities

- **Monolithic Full-Stack ISO:** csir-swa-os-1.0.0-fullstack-x86_64.iso (2.73 GB) for 100% air-gapped bare-metal.
- **Universal Online Bootstrap:** Remote curl-based bootstrap for rapid provisioning on commodity VPS nodes.
- **MicroOS Conversion Engine:** Dedicated script (convert-microos.sh) transforms vanilla openSUSE nodes.
- **VirtualBox & Hyper-V Testbeds:** Turn-key automated provisioning with 30GB disk and port forwardings.

EXECUTIVE CONCLUSION

Institutional Impact & Sovereign Technology Sovereignty

Complete Liberation from Proprietary Software Licenses with Certified National Security Compliance

CORE ARCHITECTURAL MANDATE

- Financial Sovereignty: Eliminates recurring commercial license fees for operating systems, commercial WAFs, and load balancers.
- Statutory Assurance: Built-in 1-click incident dossier generation directly fulfills national CERT-In 6-hour disclosure mandates.
- Universal Delivery: Complete deliverables suite including Monolithic DVD ISO, Web Management Console, and Full Documentation.